

Fixed backgrounds and the combined evidence

A numerical extension of the outside statistician’s review

Perspective: I approach these records as a statistician concerned with background modeling at high event counts. I take the supplied detector response, selections and signal-rate conversions as analysis assumptions. Here I evaluate the requested alternative directly: hold the estimated background fixed in each signal likelihood, and show both its nominal p -values and the outcome of repeating that procedure on complete null spectra.

What the newly computed fixed-background fits show

Omitting background covariance makes the nominal local probabilities dramatically small. Direct null scans give a different assessment: the all-three shared search has 38/256 global exceedances, and the independent-amplitude search has 36/256, on 50–100 MeV. These are modest conditional results, not discoveries with the nominal significances shown in Table 1.

Scope/domain	m_* [MeV]	Z_{nom}	p_{nom}	$k_{\text{glob}}/256$	p_{glob} [95% interval]
2015 full	51	11.906	5.48×10^{-33}	10/256	0.0391 [0.0189, 0.0707]
2016 full	87.5	8.723	1.36×10^{-18}	35/256	0.1367 [0.0971, 0.1850]
2021 10%	77	8.378	2.70×10^{-17}	96/256	0.3750 [0.3155, 0.4374]
Shared, union	22	11.360	3.31×10^{-30}	13/256	0.0508 [0.0273, 0.0853]
Shared, all three	50.5	7.800	3.10×10^{-15}	38/256	0.1484 [0.1072, 0.1980]
Free, union	51	11.597	2.13×10^{-31}	48/256	0.1875 [0.1416, 0.2408]
Free, all three	51	11.597	2.13×10^{-31}	36/256	0.1406 [0.1005, 0.1893]

Table 1: Fresh $C = 0$ results on the 0.5 MeV grid. The first probabilities are nominal local normal/chi-bar-square mappings; the last are raw k/N direct Poisson global estimates, with Clopper–Pearson intervals. Union means 19–250 MeV; all three means 50–100 MeV. At the union shared peak of 22 MeV, only 2015 contributes. The local exceedance counts for these rows are 0, 1, 0, 6, 2, 2 and 2 of 256, respectively. Add-one estimates and full precision are retained separately in the numerical ledger.

The union maximum at 22 MeV is not evidence pooled from three campaigns. The all-three shared peak at 50.5 MeV has positive 2015 and 2021 contributions, with a weak 2016 response, and lies near the 2021 search boundary; that proximity alone does not establish an artifact. The free-amplitude peak at 51 MeV is 99.7% 2015 by its contribution to q_R .

The new calculation: background fixed inside the signal likelihood

I have now computed $C = 0$ fits for 2015 full, 2016 full and 2021 10%, and two combinations. At every mass, the inherited GP recomputes the background mean b_{dj} from that spectrum’s sidebands, excluding $\pm 2.25\sigma_m$. The prediction covariance is then omitted *inside the signal likelihood*:

$$L_d(A_d; m) = \prod_{j \in W_d(m)} \text{Pois}(n_{dj} \mid b_{dj} + A_d S_{dj}(m)). \quad (1)$$

There are no background nuisance parameters. The GP arithmetic mean and original signal normalization are retained; $A = \epsilon^2/10^{-8}$.

The shared fit uses one common A . The independent-amplitude fit uses one nonnegative amplitude per active dataset at the same mass. The signed root is

$$r = \text{sign}(\hat{A}) \sqrt{2\{\log L(\hat{A}) - \log L(0)\}}.$$

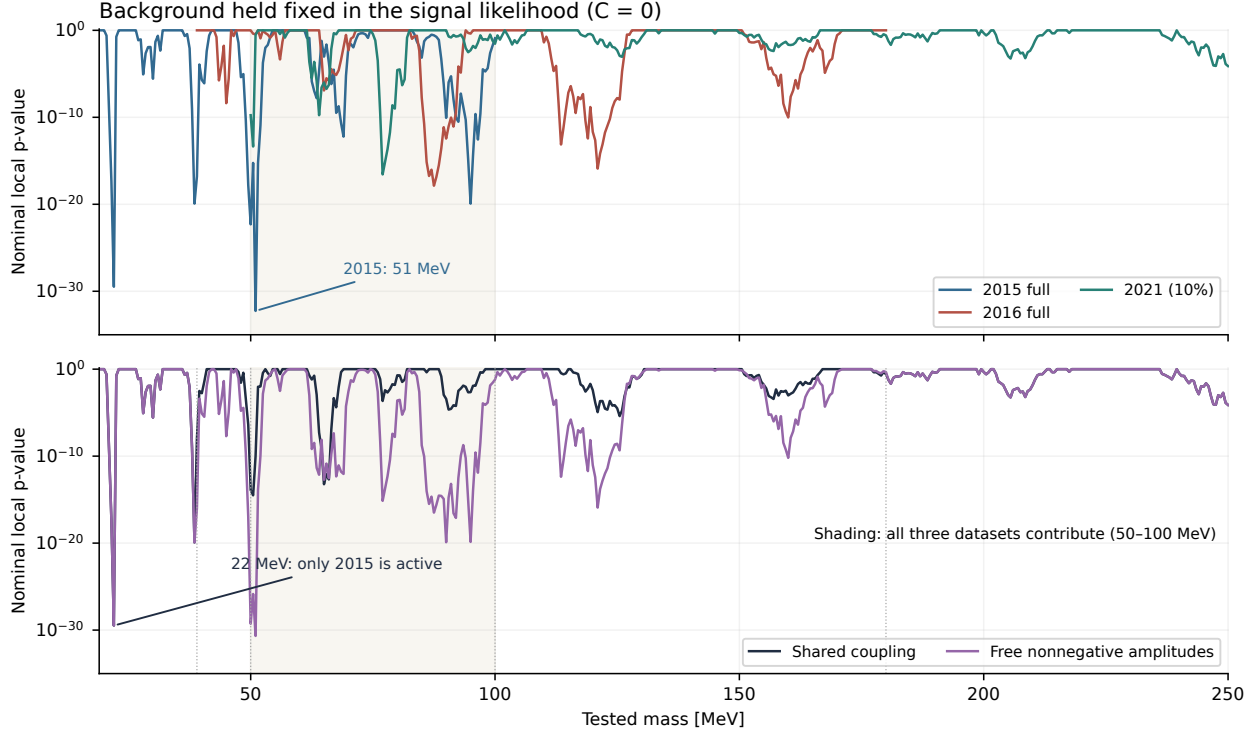


Figure 1: Computed nominal local probabilities with background covariance set to zero in the likelihood. The shaded all-three overlap and the changing union membership are explicit. Very small plotted probabilities are nominal mappings; direct null calibration is reported in Table 1.

The nominal local mapping is $p_{\text{nom}} = \bar{\Phi}(r)$ for $r > 0$. For K active independent amplitudes,

$$q_R = \sum_{d=1}^K \max(r_d, 0)^2, \quad p_{\text{nom}}(q_R) = \sum_{k=1}^K \frac{\binom{K}{k}}{2^K} P(\chi_k^2 \geq q_R), \quad q_R > 0. \quad (2)$$

This nominal chi-bar-square mixture does not identify $\sqrt{q_R}$ with Z . Zero one-sided statistics have inclusive $p = 1$. These asymptotic mappings need checking because b was estimated from the sidebands.

Individual domains are 19–100, 39–180 and 50–250 MeV. Combinations search the 19–250 MeV union with changing membership, and separately the 50–100 MeV all-three overlap. Every search uses a 0.5 MeV grid; no domain is chosen for its smaller probability.

Checking the nominal probabilities without recentering the scan

The check repeats the GP prediction and fit on 256 complete Poisson spectra from the frozen v5.8.2 sources. Each toy has its own sideband estimate, held fixed only inside its likelihood. Draws are coherent across mass and are paired with the profiled comparison. This extension uses direct Poisson scans, not a new significance-field GP approximation.

No offset is subtracted and no response width rescales the root. Global ordering uses the largest positive raw root for individual/shared fits, and $\max_m [-\log p_{\text{nom}}(q_R(m); K(m))]$ for free amplitudes. The latter accounts for changing K without reference centering. Observed and toy scans use identical statistics, grids and domains.

Local tails count exceedances at fixed mass; global tails compare search maxima. Clopper–Pearson intervals describe sampling uncertainty conditional on the fixed sources. Zero of 256 exceedances gives only the one-sided 95% upper bound $p < 0.01163$, not a resolved rare probability.

The nominated regions under this likelihood

At 66 MeV the shared root is 7.251, whereas at 92 MeV it is 3.853. The free mixture at 92 MeV gives nominal $Z = 8.521$, driven chiefly by the older samples: the individual raw roots are 6.371, 6.160 and 0.874. At 78 MeV the roots are -8.157 , -5.200 and $+7.552$; the free statistic keeps the positive 2021 component and therefore does not establish cross-campaign agreement.

Mass [MeV]	Shared p_{nom}	Local $k/256$	Free p_{nom}	Local $k/256$
66	2.07×10^{-13}	0/256	2.93×10^{-13}	3/256
78	2.06×10^{-3}	31/256	4.85×10^{-13}	3/256
90.5	2.24×10^{-5}	6/256	1.73×10^{-15}	2/256
91	2.71×10^{-5}	8/256	6.52×10^{-14}	5/256
92	5.83×10^{-5}	7/256	7.89×10^{-18}	1/256

Table 2: Actual combined fits and fixed-mass Poisson checks at the requested anchors. These counts estimate local conditional tails, not discovery probabilities after choosing a mass. Zero counts are bounds, as described above. Every individual and combined row, including global tails at these thresholds, is saved in the full CSV.

Why the nominal increase is not a significance gain

At shared 66 MeV, the toy root has mean -0.060 and SD 2.730; its deterministic root is only 0.060. The nominal 5% test rejects 59/256 null spectra (23.0%), and the nominal 1% test rejects 43/256 (16.8%). The same profiled raw-root toys have SD 1.054. This is missing background-estimation variability *before* the look-elsewhere effect, even where the mean offset is negligible. At 22 MeV, both effects matter: the null mean is 3.972 and SD is 3.743. No correction was applied to the observed roots to produce these comparisons.

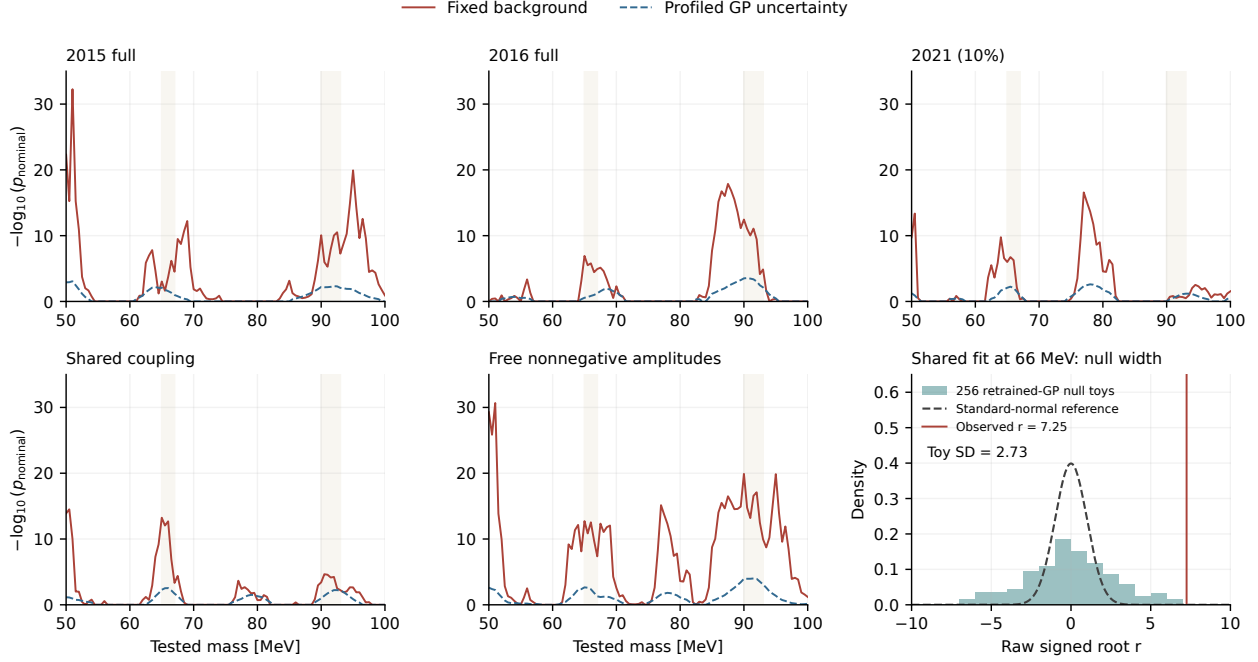


Figure 2: The first five panels compare nominal local mappings for the new $C = 0$ fits and the inherited profiled *raw* fits, with no reference centering in either. The sixth panel compares the direct shared 66 MeV null-root distribution with a standard normal. Its broad width exposes a local calibration failure before any search-multiplicity correction.

The ratio of a global probability to one of these tiny nominal local probabilities is therefore not a pure trials factor. Raw ordering remains a well-defined conditional test, but its varying null width across mass gives uneven sensitivity. For example, the 92 MeV overlap-global counts are 237/256 shared and 129/256 free, despite its small nominal tails.

My judgement is that the $C = 0$ branch is informative about model dependence, but its nominal improvement is not evidence for a stronger discovery. Using matched raw tests, the free union gives 48/256 global exceedances here versus 11/256 when profiled; the shared all-three search gives 38/256 versus 27/256. There is no uniform gain across methods or domains. Any decision to adopt $C = 0$ should compare calibrated power under specified background and signal ensembles, not select its more striking observed curve. These checks still condition on a source derived from the observed spectra; source uncertainty, source learning and prior method choices remain outside their calibration.

What I would ask of more 2021 data

My interest is in a reproducible mass, shape and rate. Removing background nuisance parameters changes the apparent ranking of the existing data, but supplies no new events. I would use the earlier profiled studies and the new fixed-background check together to design a replication, with the background policy chosen before inspecting that replication.

The earlier profiled studies made 90–93 MeV worth following for rate consistency, and 65–67 MeV worth following under a common coupling. At 92 MeV their individual fitted ϵ^2 , in units of 10^{-6} , were

$$77.48 \pm 29.96 \text{ (2015)}, \quad 9.13 \pm 2.93 \text{ (2016)}, \quad 1.65 \pm 1.35 \text{ (2021)}.$$

Those are conditional fit errors, not complete detector or normalization uncertainties. The profiled 2021 sample supplied about 82% of the local null information there while preferring the smallest amplitude. That explains why a flexible combination can look stronger than the physically restricted one. A common location does not by itself establish a common physical rate.

Use genuinely new events for the first replication test. If the later total contains the examined 10%, analyze the additional events separately before showing the pooled result. Freeze the mass hypotheses, signal widths, rate conversion, background treatment, test statistic and $\pm 2.25\sigma_m$ mask. Searching intervals around nominated candidates still carries a trials correction, as does choosing among several candidates.

At 90–93 MeV, compare the rate as well as the mass. New 2021 events could support a resolution-shaped excess at the rate predicted by a qualified common-coupling model, or make the differences among campaigns clearer. At 65–67 MeV they test the earlier shared-coupling preference. The new fixed-background preference near 51 MeV must also be judged through its dataset composition and calibration, rather than treated as an independent confirmation of the earlier studies. Around 76–80 MeV, examine the neighboring residual pattern as a background-model control.

More counts alone cannot distinguish signal from persistent model error. A fractional background error has a residual proportional to exposure, just as a persistent signal fraction does. In a counting approximation,

$$Z_{\text{mismatch}}(k) \simeq \frac{k \delta b}{\sqrt{k} b} = \sqrt{k} \frac{\delta b}{\sqrt{b}}.$$

A growing nominal score can therefore reveal a modeling error more precisely. Check fractional residuals, peak width, sidebands and justified control selections, including possible signal absorption when constructing the generating source. The v5.0.5 full-exposure projections assume persistence of selected partial-sample excesses; they are scenarios, not forecasts of unseen data.

Evidence and scope. This extension uses the portable v5.8.2 spectra, frozen nominal generating means, complete toy bank and raw profiled fields; the original review used v5.0.5 Sections 4.20/6.11, its calibration and projection material, and v5.8.4/v5.8.5 combination studies. The source archive includes numerical curves, toy fields, hashes, the statistician's assessment and independent likelihood checks. All empirical probabilities here remain conditional on the frozen source and inherited analysis choices. The original reports remain unchanged.